Enterprise vSphere Environment Build and Configuration Portfolio

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Environment: LAB2 (vSphere 8 / ESXi)

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1.Virtual Machine Provisioning and Deployment

Objective: Deploy a multi-tier application architecture with role-specific resource and storage allocation.

Deployment Method: All VMs were created using the vSphere Client "New Virtual Machine" wizard and deployed to the LAB2 cluster. Storage provisioning was specifically chosen based on workload requirements.

- **Web Tier (Web-01, Web-02):** 2 vCPUs, 4 GB RAM, 40 GB Thin Provisioned storage, connected to PG-Web-Tier (VLAN 10).
- **Application Tier (App-01):** 4 vCPUs, 8 GB RAM, 40 GB Thin Provisioned storage, connected to PG-Web-Tier (VLAN 10).
- **Database Tier (DB-1):** 4 vCPUs, 8 GB RAM, 100 GB Thick Eager Zeroed storage, isolated on PG-DB-Tier (VLAN 20).

Storage Strategy: Thin Provisioning was used for the Web and App tiers to conserve storage space. **Thick Provision Eager Zeroed** was used for the Database server (DB-1). This is a VMware best practice for database workloads because it pre-allocates and zeroes out all blocks during creation, eliminating lazy-zeroing delays and ensuring consistent, high-performance disk I/O.

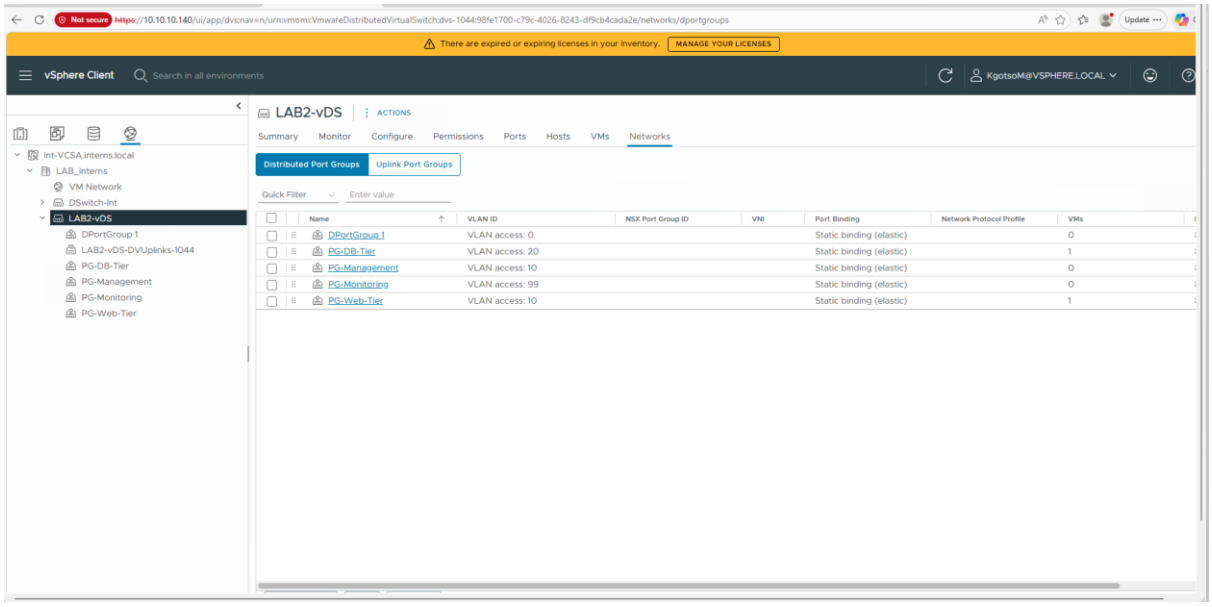
2.Distributed Switch & Network Segmentation

Objective: Centralize network management and enforce security boundaries using VLANs.

Action Taken:

- Created a new Distributed Switch named **LAB2-vDS** (Version 7.0.0 to ensure ESXi host compatibility).
- Created the following Distributed Port Groups to segment traffic:
    - **PG-Web-Tier** (VLAN 10)
    - **PG-DB-Tier** (VLAN 20)
    - **PG-Management** (VLAN 10)
    - **PG-Monitoring** (VLAN 99)

Evidence:



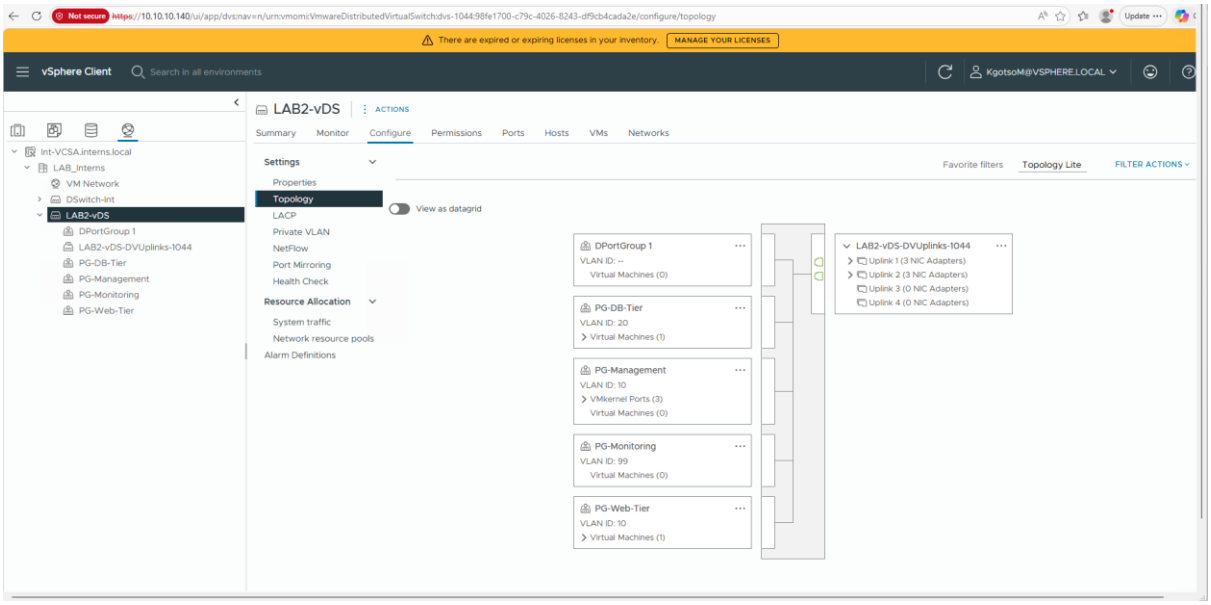
3.Host Migration & Physical Uplinks

Objective: Connect physical ESXi hosts to the Distributed Switch without causing management network downtime.

Action Taken:

- Added LAB2 ESXi hosts (10.10.10.143, 144, 145, 146) to the LAB2-vDS.
- Assigned physical adapters: vmnic0 to **Uplink 1** and vmnic1 to **Uplink 2**.
- Migrated the management VMkernel adapter (vmk0) to the newly created **PG-Management** port group to prevent loss of connectivity.

Evidence:



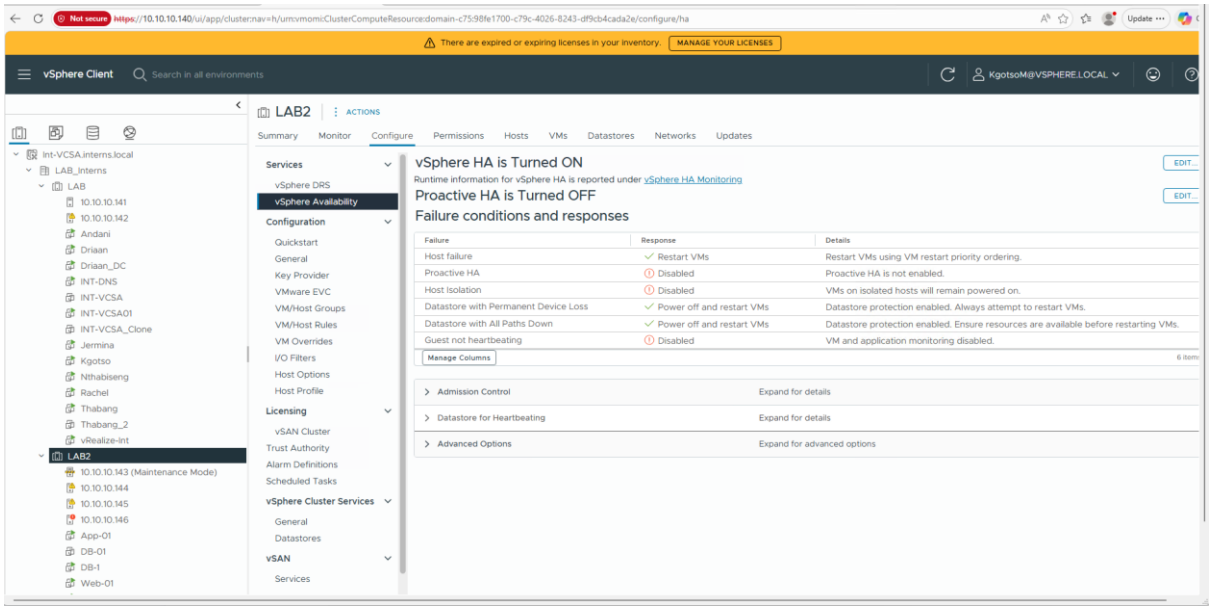
4.Cluster Configuration: DRS, HA, and Affinity Rules

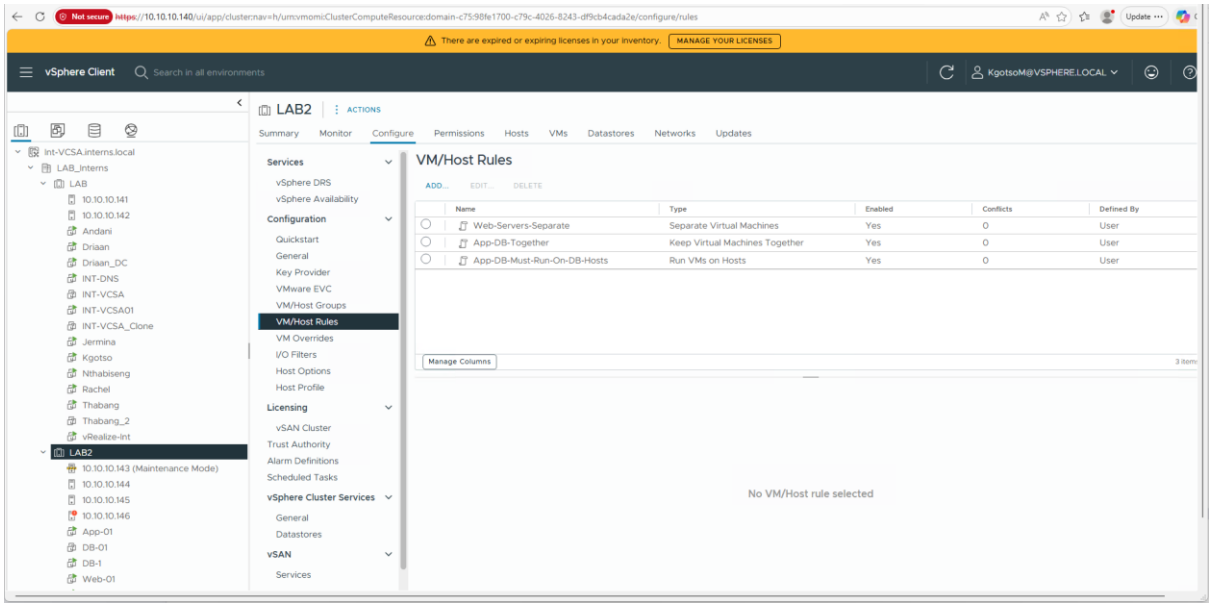
Objective: Optimize resource utilization, ensure high availability, and enforce VM placement policies.

Action Taken:

- **vSphere HA:** Enabled on the LAB2 cluster to automatically restart VMs on surviving hosts in the event of a physical host failure.
- **DRS (Distributed Resource Scheduler):** Enabled to automatically balance compute workloads across the 4 ESXi hosts**.**
- **VM/Host Groups & Rules:** Created Anti-Affinity rules to ensure critical VMs (like Web-01 and Web-02) are placed on separate physical hosts to prevent a single point of failure.

Evidence:





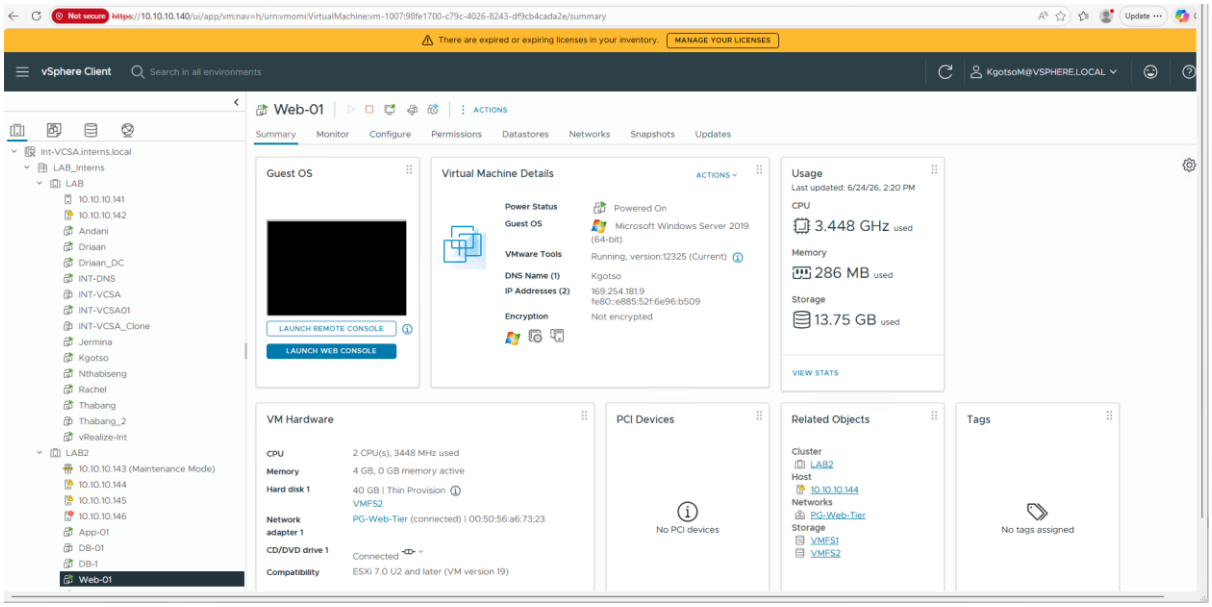
5.Workload Migration

Objective: Move virtual machines from the legacy Standard Switch to the secure Distributed Switch port groups.

Action Taken:

- Migrated **Web-01, Web-02, and App-01** network adapters to **PG-Web-Tier**.
- Migrated **DB-1** network adapter to **PG-DB-Tier**.

Evidence:



6.Network Monitoring (Port Mirroring)

Objective: Implement passive network traffic analysis for security and troubleshooting.

Action Taken:

- Created a Distributed Port Mirroring session named **Web01-Monitoring-Session**.
- Set **Web-01** as the **Source** (to monitor incoming/outgoing web traffic at the perimeter).
- Set **DB-1** as the **Destination** (to receive the mirrored traffic for analysis).
- Set session status to **Enabled**

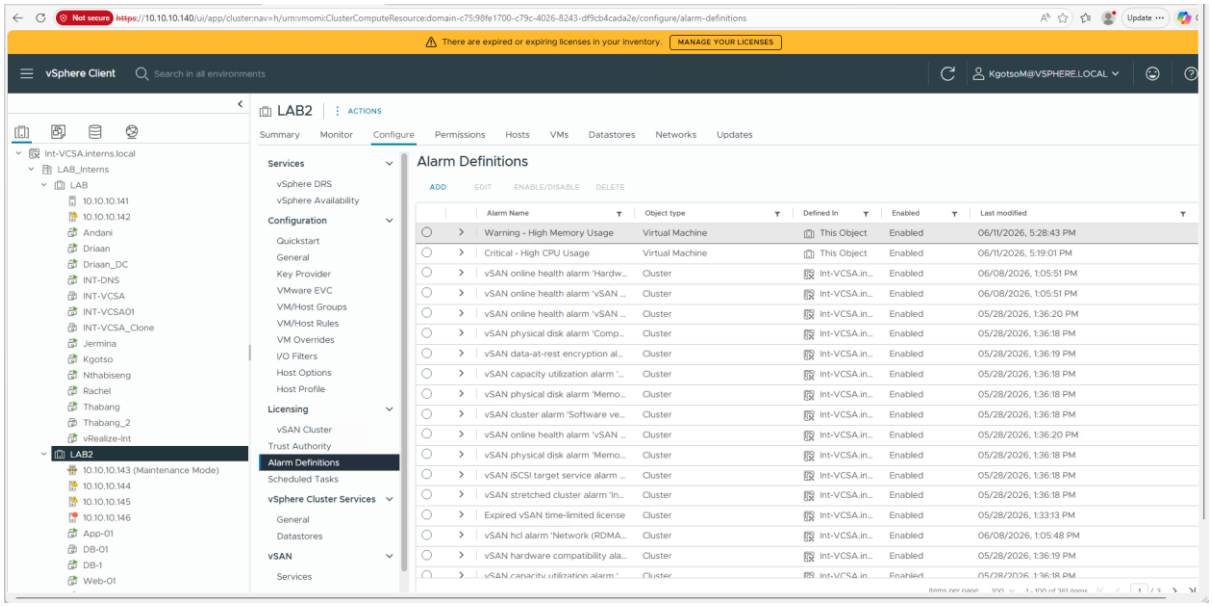
7.Alarm Configuration and Resolution

Objective: Monitor, diagnose, and resolve infrastructure alerts to ensure operational health.

Action Taken:

- Configured Alarm Definitions for **High CPU Usage** and **High Memory Usage** on the cluster to proactively detect performance bottlenecks.
- Resolved critical infrastructure alerts (Network Connectivity Lost, Uplink Redundancy Lost) generated during the initial migration phase.
- Performed "Reset to Green" on all alarms to confirm the environment was fully healthy and operational.

Evidence:



8.Troubleshooting & Problem Resolution

During the implementation, several technical challenges were encountered and successfully resolved:

1. **vDS Version Incompatibility:** The initial vDS version was incompatible with the ESXi hosts.
    - *Resolution:* Recreated the LAB2-vDS using version 7.0.0, which matched the host capabilities.
2. **Management Network Isolation:** During the initial host migration, the hosts lost connection to vCenter because the management traffic was placed on the default VLAN 0 instead of the required VLAN 10.
    - *Resolution:* Identified the correct Management VLAN (10), created the PG-Management port group, and successfully re-migrated the vmk0 adapters.

9.Advanced Networking - LACP Configuration

Objective: Configure Link Aggregation Control Protocol (LACP) to increase network bandwidth and provide uplink redundancy.

Action Taken:

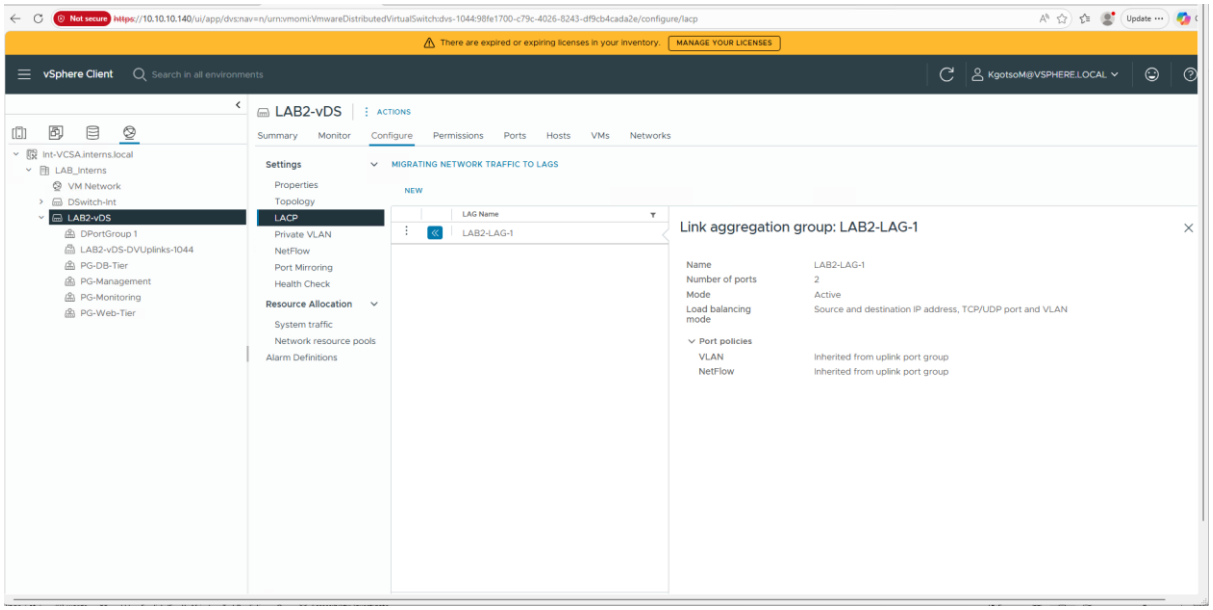
- Navigated to **LAB2-vDS** > **Configure** > **LACP**.

- Created a new Link Aggregation Group (LAG) named **LAB2-LAG-1**.
- Configured the following settings:
    - **Number of Ports:** 2 (to bundle two physical network adapters)
    - **Mode:** Active (enables dynamic LACP negotiation with the physical switch)
    - **Load Balancing:** Source and Destination IP Address, TCP/UDP Port (ensures optimal traffic distribution while maintaining packet order)
- The LAG is now ready to accept physical uplinks for aggregation.

Benefits:

- **Increased Bandwidth:** Two physical adapters work together as a single logical connection, providing double the throughput
- **Automatic Failover:** If one physical cable fails, traffic automatically continues through the remaining cable with zero downtime
- **Load Balancing:** Network traffic is intelligently distributed across both physical adapters for optimal performance

Evidence:



10.Network Traffic Monitoring - NetFlow Configuration

Objective: Implement network traffic monitoring to analyze bandwidth usage and identify traffic patterns.

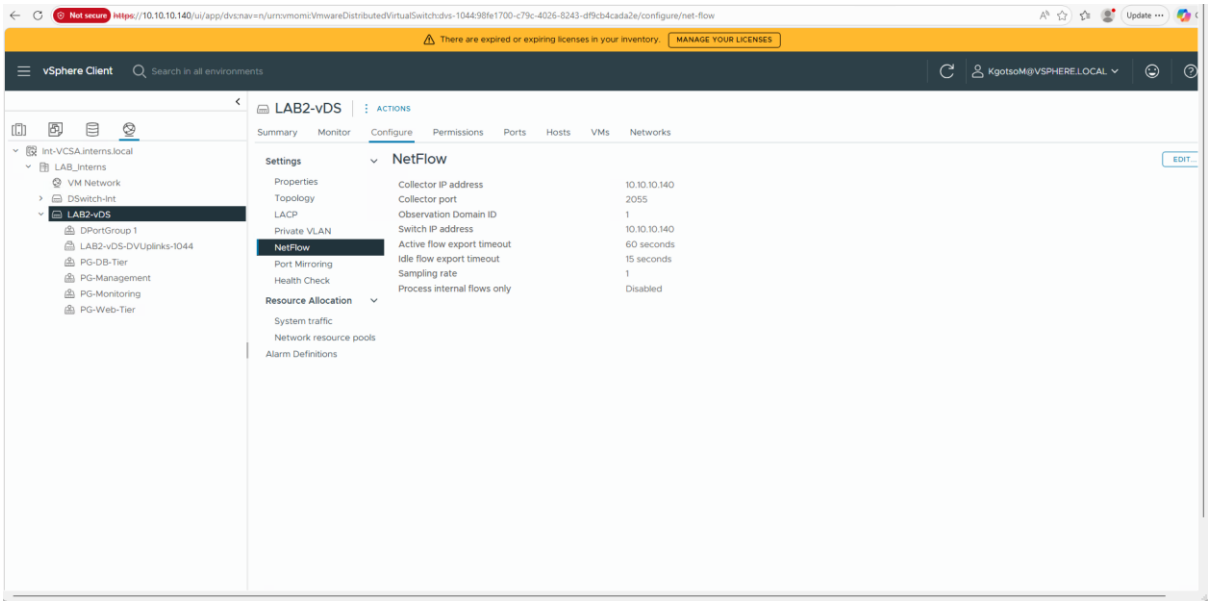
Action Taken:

- Configured NetFlow on the LAB2-vDS to monitor all network traffic
- Set the NetFlow collector to 10.10.10.140 on port 2055
- Configured sampling rate to 1 (capture all flows) for complete visibility
- Set active flow timeout to 60 seconds and idle flow timeout to 15 seconds
- Enabled monitoring of both internal and external flows

Benefits:

- **Traffic Analysis:** Identify which VMs are consuming the most bandwidth
- **Security Monitoring:** Detect unusual traffic patterns that may indicate security breaches
- **Performance Troubleshooting:** Analyze network flows to identify bottlenecks
- **Capacity Planning:** Understand traffic patterns to plan for future network upgrades

Evidence:



11.Network Resource Prioritization - NIOC Configuration

Objective: Configure Network I/O Control (NIOC) to ensure critical infrastructure traffic receives priority during network congestion.

Action Taken:

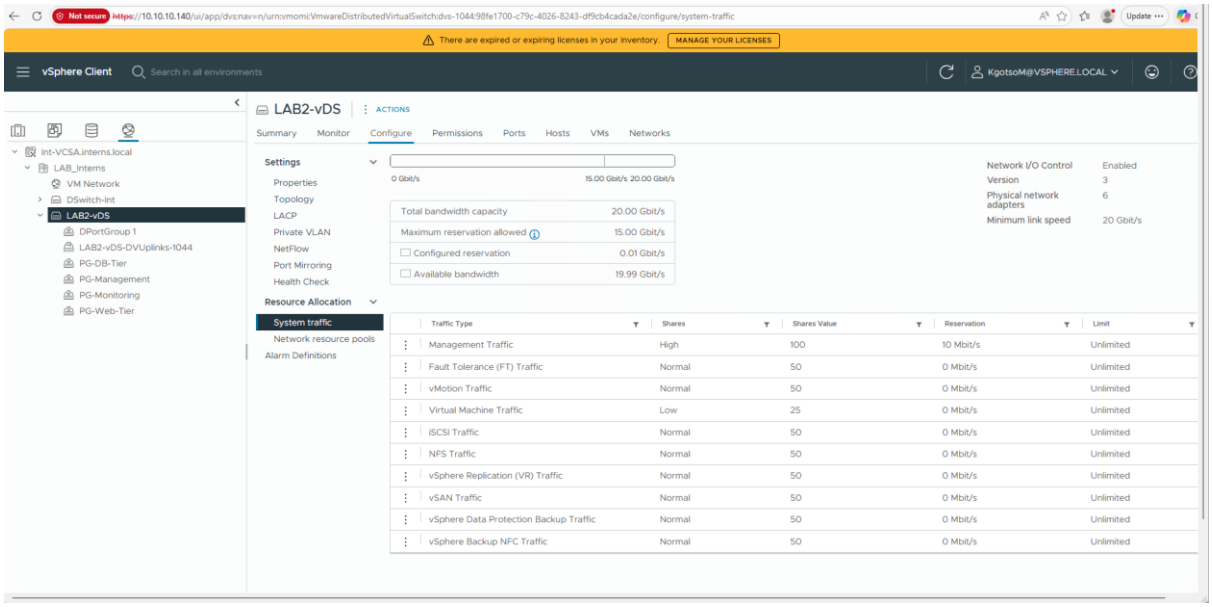
- Verified NIOC Version 3 is enabled on LAB2-vDS
- Configured traffic prioritization using shares and reservations:
    - **Management Traffic:** High shares (100) with 10 Mbit/s reservation
    - **Virtual Machine Traffic:** Low shares (25) to prevent resource starvation
    - **vMotion, Storage, and other traffic:** Normal shares (50) for balanced performance
- Total bandwidth capacity: 20 Gbit/s with 15 Gbit/s maximum reservation allowed

Benefits:

- **Prevents Network Congestion:** Critical traffic always gets bandwidth first
- **Ensures Management Connectivity:** Host management never loses connectivity
- **Optimizes Performance:** Traffic is prioritized based on business criticality

- **Guarantees SLA Compliance:** Reservations ensure minimum bandwidth for critical services

Evidence:



12.Network Resource Pools

Objective: Create custom bandwidth allocations for specific application tiers to ensure Service Level Agreements (SLAs) are met.

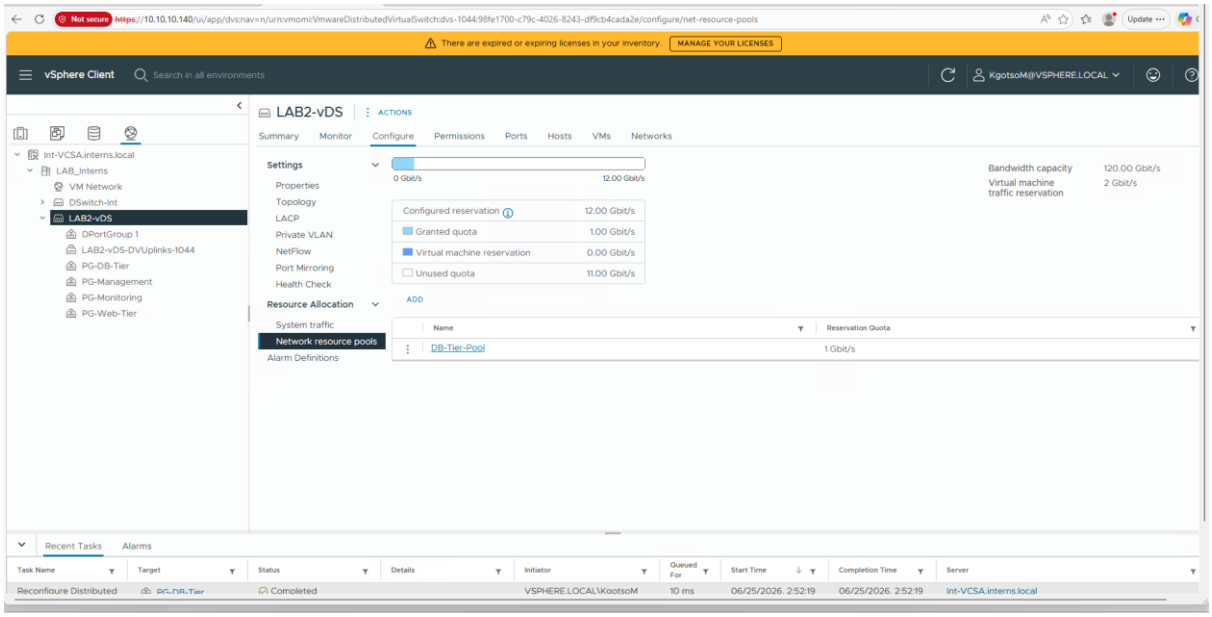
Action Taken:

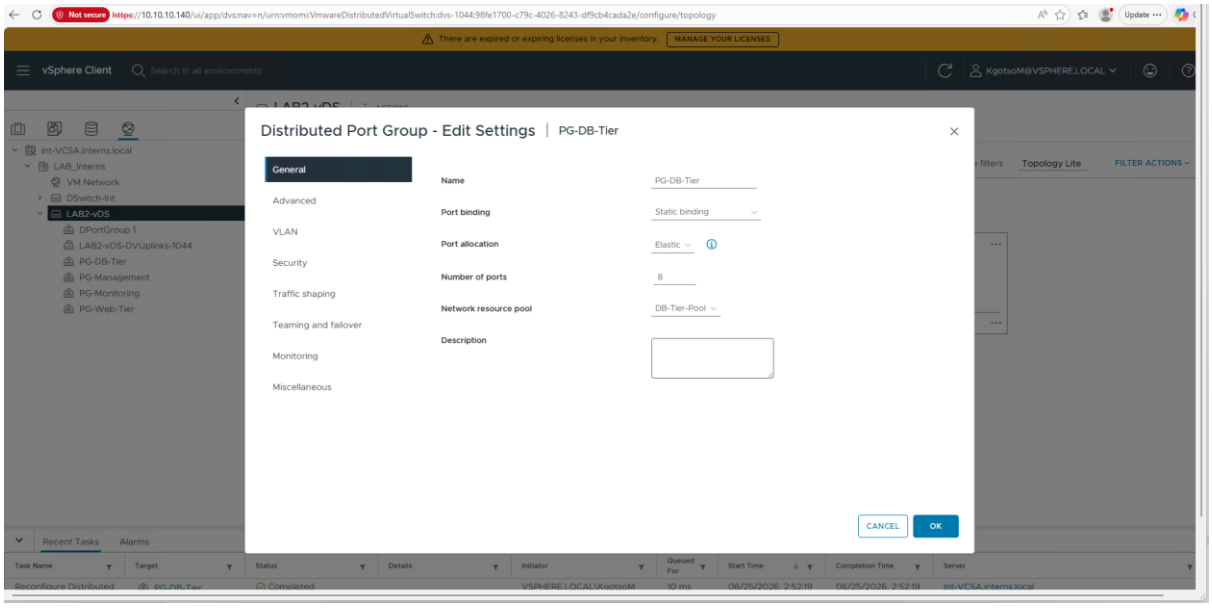
- Reserved 2000 Mbit/s of bandwidth for Virtual Machine system traffic to enable resource pooling.
- Created a custom Network Resource Pool named **DB-Tier-Pool** with a 1000 Mbit/s reservation quota.
- Assigned the **PG-DB-Tier** port group to the DB-Tier-Pool.

Benefits:

- **Guaranteed Performance:** Database traffic is guaranteed 1 Gbps of bandwidth regardless of other network activity.
- **Quality of Service (QoS):** Prevents heavy web traffic or backups from starving the database of network resources.

Evidence:





13.vMotion Configuration and Live Migration Testing

Objective: Enable live virtual machine migration capabilities to allow workload mobility, zero-downtime maintenance, and dynamic resource balancing across the ESXi cluster.

Configuration Steps:

Step 1: Enable vMotion on VMkernel Adapters

- Navigated to each ESXi host (10.10.10.144, 10.10.10.145, 10.10.10.146) in the LAB2 cluster
- Accessed **Configure** > **Networking** > **VMkernel adapters**

- Edited the **vmk0** adapter and enabled the **vMotion** service
- Verified that both **Management** and **vMotion** services were enabled on each host
- This configuration allows hosts to communicate and transfer VM memory/state during live migrations

Step 2: Perform Live vMotion Migration Test

- Right-clicked **Web-01** virtual machine and selected **Migrate**
- Selected **Change compute resource only** migration type
- Chose a different ESXi host as the destination (migrated from 10.10.10.144 to 10.10.10.145)
- Verified network mapping (PG-Web-Tier was automatically mapped due to Distributed Switch configuration)
- Set vMotion priority to **High**
- Completed the migration wizard and initiated the live migration

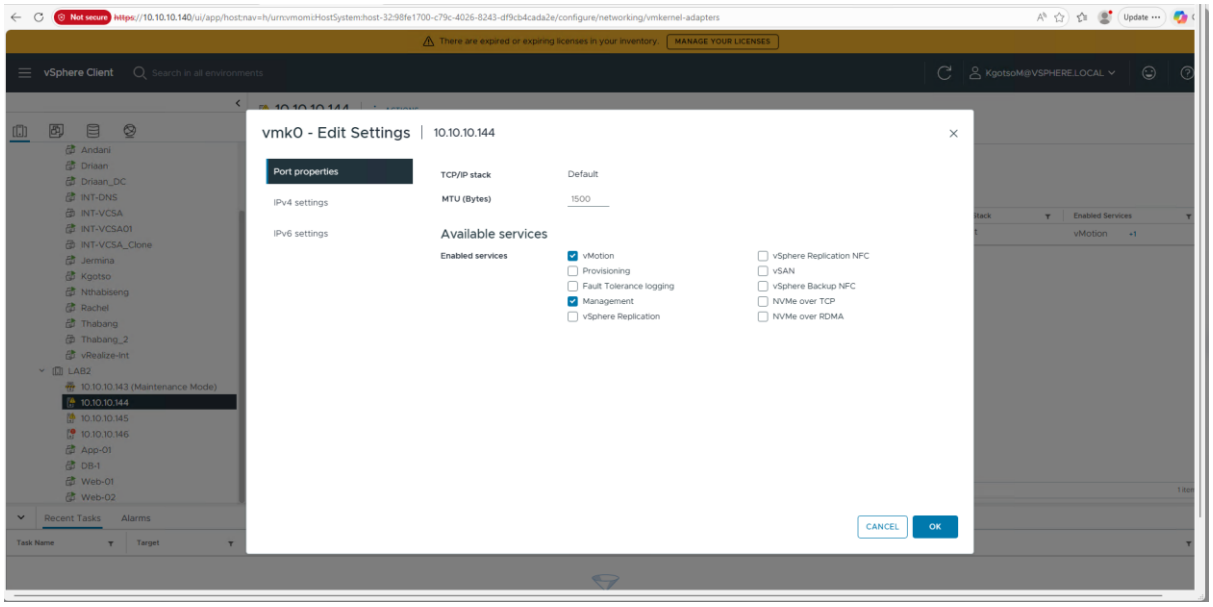
Step 3: Validate Migration Success

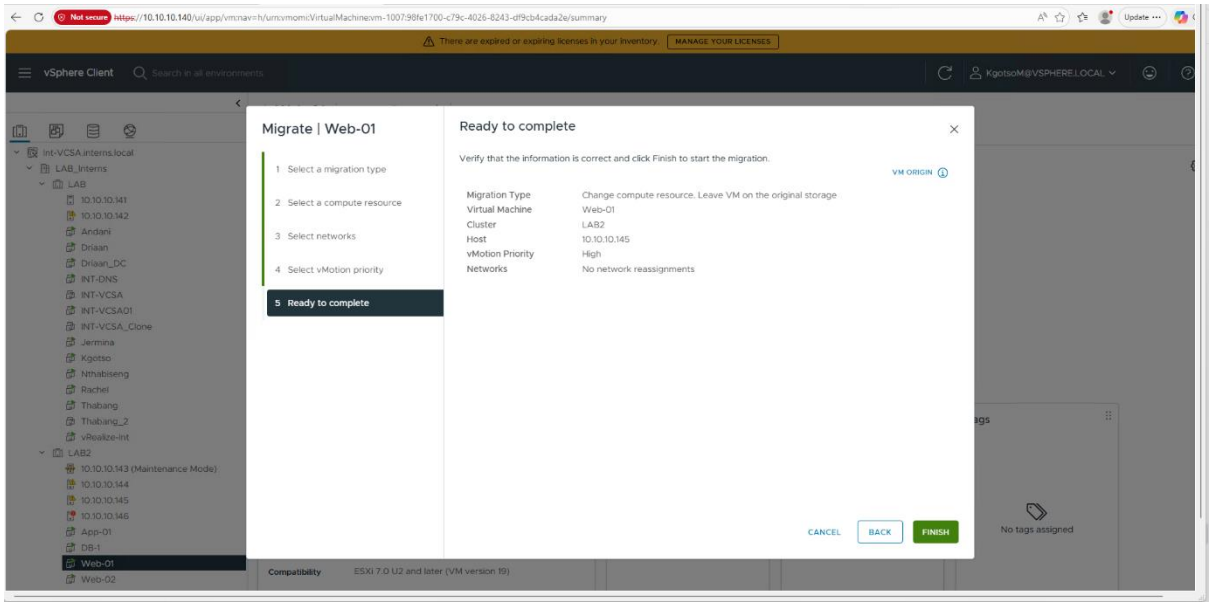
- Monitored the "Recent Tasks" pane for migration progress
- Confirmed successful completion with **"Relocate virtual machine"** status showing **Completed**
- Verified Web-01 was running on the new host with zero downtime
- Confirmed network connectivity was maintained throughout the migration

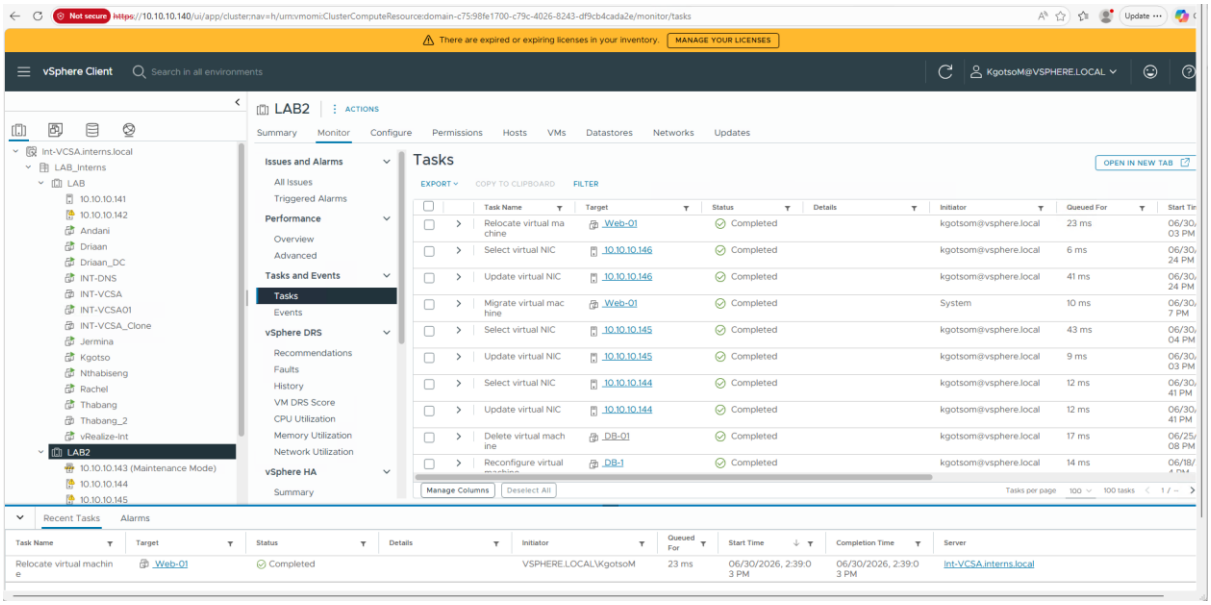
Benefits:

- **Zero-Downtime Maintenance:** VMs can be moved off hosts before maintenance or patching without service interruption
- **Workload Mobility:** Enables manual or automated (DRS) load balancing across the cluster
- **Disaster Avoidance:** Proactively move VMs away from failing hardware before outages occur
- **Operational Flexibility:** Perform host updates, hardware upgrades, and resource optimization without impacting end users
- **Seamless Networking:** Distributed Switch ensures network connectivity is maintained throughout migration

Evidence:


vmk0 - Edit Settings | 10.10.10.144
Port properties
IPv4 settings
IPv6 settings
TCP/IP stack
Default
MTU (Bytes)
1500
Available services
Enabled services
vMotion
Provisioning
Fault Tolerance logging
Management
vSphere Replication
vSphere Replication NFC
vSAN
vSphere Backup NFC
NVMe over TCP
NVMe over RDMA
CANCEL
OK


Migrate | Web-01
1 Select a migration type
2 Select a compute resource
3 Select networks
4 Select vMotion priority
5 Ready to complete
Ready to complete
Verify that the information is correct and click Finish to start the migration.
VM ORIGIN
Migration Type
Change compute resource. Leave VM on the original storage
Virtual Machine
Web-01
Cluster
LAB2
Host
10.10.10.145
vMotion Priority
High
Networks
No network reassignments
CANCEL
BACK
FINISH



14.Distributed Switch Health Check Configuration

Objective: Enable automated network validation to proactively detect physical network misconfigurations before they cause outages.

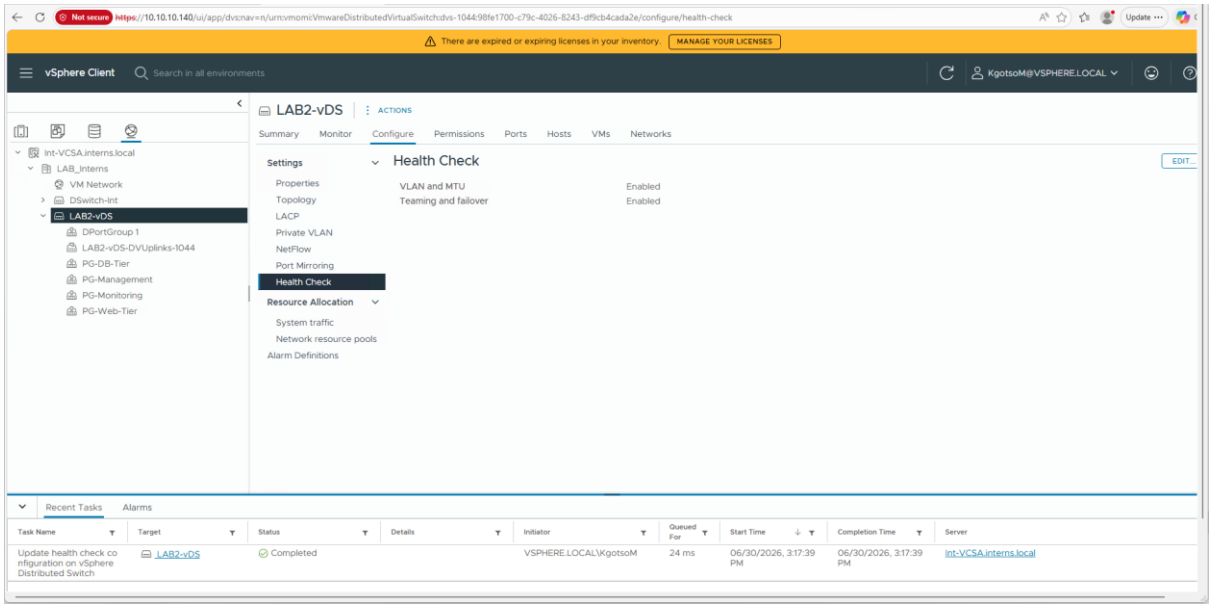
Action Taken:

- Navigated to **LAB2-vDS** > **Configure** > **Health Check**.
- Enabled the **VLAN and MTU** health check to verify that physical switch ports match the virtual switch configuration.
- Enabled the **Teaming and failover** health check to ensure physical uplinks are correctly connected and functioning.
- Verified the configuration task completed successfully in the Recent Tasks pane.

Benefits:

- **Proactive Monitoring:** Automatically detects misconfigurations (like a wrong VLAN tag on a physical switch port) without waiting for a user to report an issue.
- **Prevents Outages:** Catches MTU (packet size) mismatches that can cause silent network drops or performance issues.
- **Operational Efficiency:** Reduces troubleshooting time by pinpointing exactly which host or uplink has a configuration mismatch.

Evidence:



15.Content Library Configuration

Objective: Implement a centralized Content Library for standardized VM template management and rapid deployment.

Action Taken:

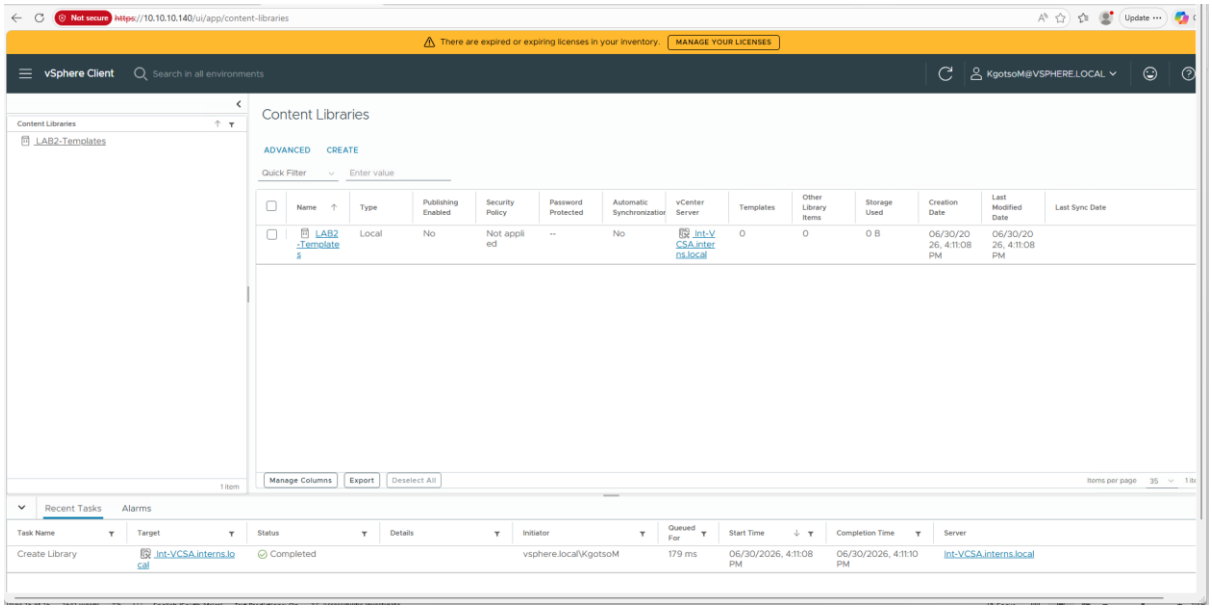
- Navigated to **Menu** > **Content Libraries** and created a new **Local Content Library**
- Named the library **LAB2-Templates**
- Configured the library to use the **VMFS2** shared datastore for centralized storage
- Left security policy disabled for lab environment flexibility
- Library is now ready to store VM templates, ISO images, and OVF/OVA packages for rapid deployment

Benefits:

- **Rapid Deployment:** Deploy pre-configured VM templates in minutes instead of building from scratch
- **Standardization:** Ensures all VMs are deployed with consistent configurations and security settings
- **Centralized Management:** Single source of truth for templates accessible across the entire cluster
- **Efficiency:** Reduces manual configuration errors and administrative overhead

- **Scalability:** Can be subscribed to by other vCenter instances for multi-site consistency

Evidence:



16.Content Library Configuration and Template Management

Objective: Implement a centralized Content Library to standardize virtual machine deployments, ensure configuration consistency, and enable rapid provisioning.

Action Taken:

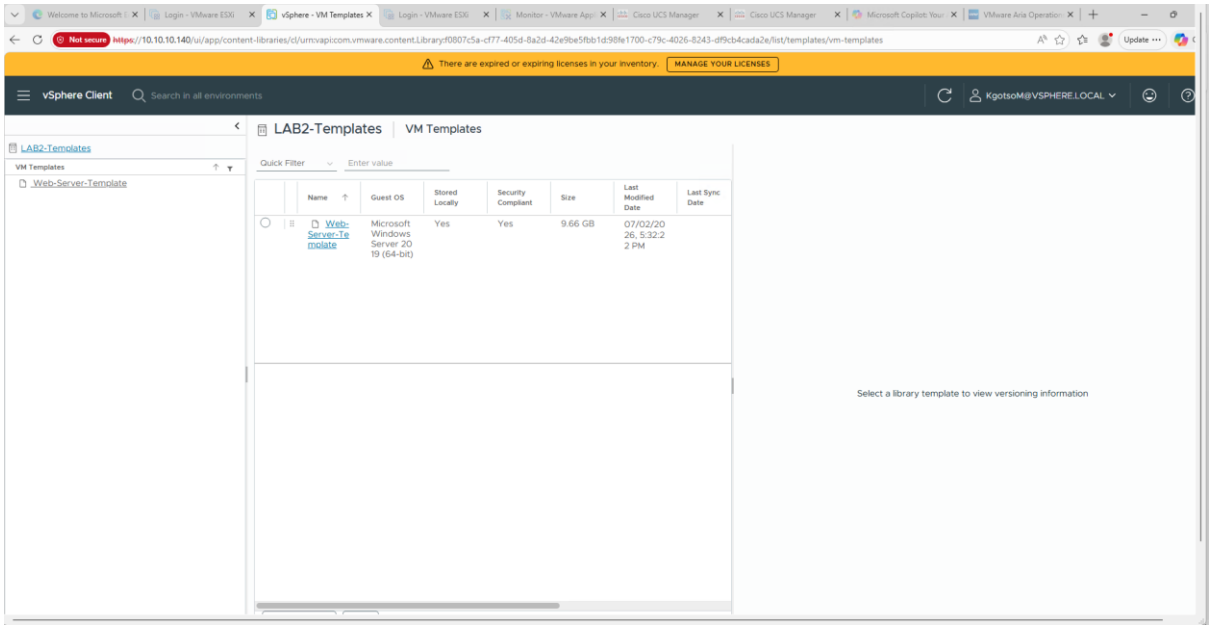
- Created a **Local Content Library** named LAB2-Templates backed by the shared VMFS2 datastore.
- Encountered a "Failed to quiesce" error during the initial clone attempt due to the source VM (Web-02) being powered on.
- **Troubleshooting:** Resolved the issue by gracefully shutting down the guest OS of Web-02 to ensure a clean file system state before retrying the clone operation.
- Successfully cloned Web-02 as Web-Server-Template and published it to the Content Library.
- Verified the template is stored locally and is security compliant.

Benefits:

- **Standardization:** Ensures all newly deployed web servers have identical hardware, OS, and baseline configurations.

- **Rapid Provisioning:** Reduces VM deployment time from hours (manual OS installation) to minutes (template deployment).
- **Centralized Management:** Provides a single source of truth for templates across the vCenter environment.
- **Error Reduction:** Eliminates manual configuration drift and human error during the VM build process.

Evidence:



17.Rapid VM Deployment from Content Library Template

Objective: Demonstrate the ability to rapidly provision standardized virtual machines using Content Library templates.

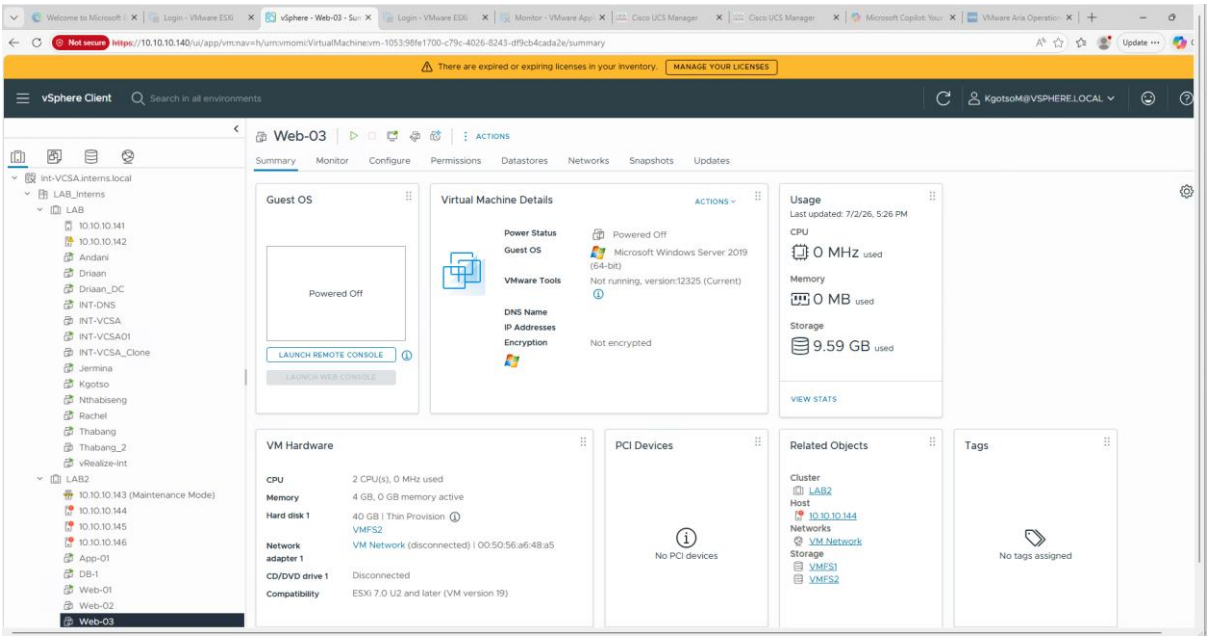
Action Taken:

- Deployed **Web-03** from the Web-Server-Template stored in the LAB2-Templates Content Library
- Selected the LAB2 cluster as the compute resource and VMFS2 as the shared datastore
- Deployment completed in under 2 minutes using the template's existing configuration
- VM deployed with the same network settings as the source template

Benefits:

- **Speed:** Deployed a fully configured Windows Server 2019 VM in minutes instead of hours
- **Consistency:** Web-03 maintains the same baseline configuration as the template
- **Zero Configuration Errors:** All settings (CPU, RAM, storage) were pre-configured in the template
- **Scalability:** This same process can be repeated to deploy dozens of VMs with consistent configurations

Evidence:



18.Infrastructure Automation with PowerCLI

Objective: Transition from manual GUI deployments to automated Infrastructure as Code (IaC) provisioning to scale the environment rapidly and consistently.

Action Taken:

- Installed and configured **VMware PowerCLI** on a management workstation.
- Authored a PowerShell script to automate the deployment of 5 virtual machines (Web-Auto-04 through Web-Auto-08).
- The script programmatically connected to vCenter, retrieved the Web-Server-Template from the **LAB2-Templates Content Library**, and deployed the VMs to the LAB2 cluster and VMFS2 datastore using Thin Provisioning.
- Successfully deployed all 5 VMs in under 2 minutes with zero manual GUI interaction.

Benefits:

- **Speed & Scale:** Reduced deployment time from ~15 minutes per VM (manual) to seconds per VM (automated).
- **Consistency:** Eliminated human error; every VM was deployed with the exact same hardware and network configuration inherited from the template.
- **Repeatability:** The script can be saved, version-controlled, and reused to deploy hundreds of VMs for future projects or disaster recovery.

Evidence:

